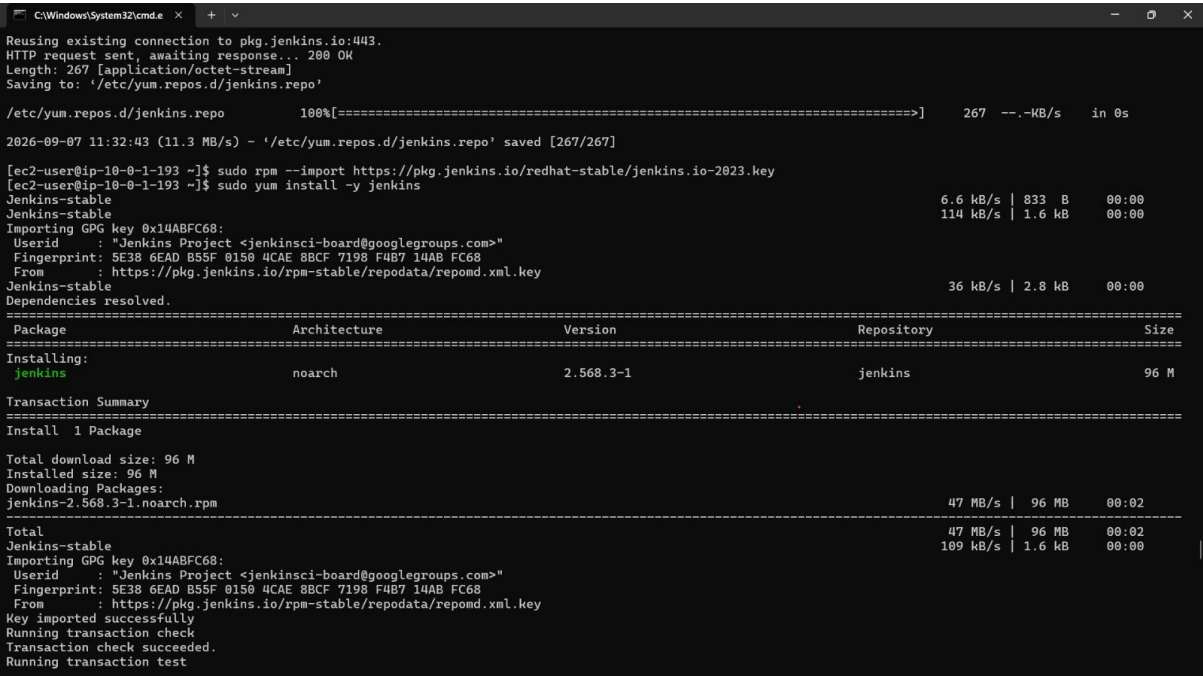


1. Browser showing the app running on localhost:3000 ("Hello from my DevOps App! CI/CD pipeline working")



2. Manual Jenkins/wget install steps in the terminal, SSH key listings




```
C:\Windows\System32\cmd.exe x + v

+ capacity_reservation_specification (known after apply)

+ cpu_options (known after apply)

+ ebs_block_device (known after apply)

+ enclave_options (known after apply)

+ ephemeral_block_device (known after apply)

+ instance_market_options (known after apply)

+ maintenance_options (known after apply)

+ metadata_options (known after apply)

+ network_interface (known after apply)

+ private_dns_name_options (known after apply)

+ root_block_device {
+   delete_on_termination = true
+   device_name            = (known after apply)
+   encrypted              = (known after apply)
+   iops                   = (known after apply)
+   kms_key_id             = (known after apply)
+   tags_all               = (known after apply)
+   throughput             = (known after apply)
+   volume_id              = (known after apply)
+   volume_size            = 30
+   volume_type            = "gp3"
}

Plan: 1 to add, 0 to change, 0 to destroy.

Changes to Outputs:
+ jenkins_public_ip = (known after apply)
+ jenkins_url       = (known after apply)

Do you want to perform these actions?
Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.
```

```
C:\Windows\System32\cmd.exe x + v

+ ebs_optimized                = (known after apply)
+ enable_primary_ipv6          = (known after apply)
+ get_password_data            = false
+ host_id                     = (known after apply)
+ host_resource_group_arn      = (known after apply)
+ iam_instance_profile         = "jenkins-eks-demo-jenkins-profile"
+ id                           = (known after apply)
+ instance_initiated_shutdown_behavior = (known after apply)
+ instance_lifecycle           = (known after apply)
+ instance_state               = (known after apply)
+ instance_type               = "t3.micro"
+ ipv6_address_count           = (known after apply)
+ ipv6_addresses               = (known after apply)
+ key_name                     = "jenkins-1"
+ monitoring                   = (known after apply)
+ outpost_arn                  = (known after apply)
+ password_data                = (known after apply)
+ placement_group              = (known after apply)
+ placement_partition_number   = (known after apply)
+ primary_network_interface_id = (known after apply)
+ private_dns                  = (known after apply)
+ private_ip                   = (known after apply)
+ public_dns                   = (known after apply)
+ public_ip                    = (known after apply)
+ secondary_private_ips        = (known after apply)
+ security_groups              = (known after apply)
+ source_dest_check            = true
+ spot_instance_request_id     = (known after apply)
+ subnet_id                    = "subnet-0e66aae71c4614d5b"
+ tags                         = {
+   "Name" = "jenkins-eks-demo-jenkins-server"
}
+ tags_all                     = {
+   "Name" = "jenkins-eks-demo-jenkins-server"
}
+ tenancy                      = (known after apply)
+ user_data                    = "452e1b08b09d986984eaa1415cc61af16a54ed8"
+ user_data_base64             = (known after apply)
+ user_data_replace_on_change = false
+ vpc_security_group_ids       = [
+   "sg-086058265cbd4acaf",
]

+ capacity_reservation_specification (known after apply)
```

```
C:\Windows\System32\cmd.exe x + v
+ enable_dns_support = true
+ enable_network_address_usage_metrics = (known after apply)
+ id = (known after apply)
+ instance_tenancy = "default"
+ ipv6_association_id = (known after apply)
+ ipv6_cidr_block = (known after apply)
+ ipv6_cidr_block_network_border_group = (known after apply)
+ main_route_table_id = (known after apply)
+ owner_id = (known after apply)
+ tags = {
+   + "Name" = "jenkins-eks-demo-vpc"
+ }
+ tags_all = {
+   + "Name" = "jenkins-eks-demo-vpc"
+ }
}

Plan: 10 to add, 0 to change, 0 to destroy.

Changes to Outputs:
+ jenkins_public_ip = (known after apply)
+ jenkins_url = (known after apply)
+ vpc_id = (known after apply)

Do you want to perform these actions?
Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value: yes

aws_iam_role.jenkins_role: Creating...
aws_vpc.main: Creating...
aws_iam_role.jenkins_role: Creation complete after 2s [id=jenkins-eks-demo-jenkins-role]
aws_iam_role_policy_attachment.jenkins_admin: Creating...
aws_iam_instance_profile.jenkins_profile: Creating...
aws_iam_role_policy_attachment.jenkins_admin: Creation complete after 1s [id=jenkins-eks-demo-jenkins-role-202609071109542734000000001]
aws_vpc.main: Still creating... [00m10s elapsed]
aws_iam_instance_profile.jenkins_profile: Creation complete after 8s [id=jenkins-eks-demo-jenkins-profile]
aws_vpc.main: Creation complete after 12s [id=vpc-0d55b04ec799793e8]
aws_internet_gateway.main: Creating...
aws_subnet.public: Creating...
aws_security_group.jenkins: Creating...
aws_internet_gateway.main: Creation complete after 0s [id=igw-0015e55ffc6fddb2c]
aws_route_table.public: Creating...

+ "Name" = "jenkins-eks-demo-jenkins-sg"
+ tags_all = {
+   + "Name" = "jenkins-eks-demo-jenkins-sg"
+ }
+ vpc_id = (known after apply)
}

# aws_subnet.public will be created
+ resource "aws_subnet" "public" {
+   arn = (known after apply)
+   assign_ipv6_address_on_creation = false
+   availability_zone = "ap-south-1a"
+   availability_zone_id = (known after apply)
+   cidr_block = "10.0.1.0/24"
+   enable_dns64 = false
+   enable_resource_name_dns_a_record_on_launch = false
+   enable_resource_name_dns_aaaa_record_on_launch = false
+   id = (known after apply)
+   ipv6_cidr_block_association_id = (known after apply)
+   ipv6_native = false
+   map_public_ip_on_launch = true
+   owner_id = (known after apply)
+   private_dns_hostname_type_on_launch = (known after apply)
+   tags = {
+     + "Name" = "jenkins-eks-demo-public-subnet"
+   }
+   tags_all = {
+     + "Name" = "jenkins-eks-demo-public-subnet"
+   }
+   vpc_id = (known after apply)
}

# aws_vpc.main will be created
+ resource "aws_vpc" "main" {
+   arn = (known after apply)
+   cidr_block = "10.0.0.0/16"
+   default_network_acl_id = (known after apply)
+   default_route_table_id = (known after apply)
+   default_security_group_id = (known after apply)
+   dhcp_options_id = (known after apply)
+   enable_dns_hostnames = true
+   enable_dns_support = true
+   enable_network_address_usage_metrics = (known after apply)
```

```

+ self = false
+ to_port = 0
# (1 unchanged attribute hidden)
},
+ id = (known after apply)
+ ingress = [
+ {
+ cidr_blocks = [
+ "59.92.237.185/32",
+ ]
+ description = "Jenkins UI"
+ from_port = 8080
+ ipv6_cidr_blocks = []
+ prefix_list_ids = []
+ protocol = "tcp"
+ security_groups = []
+ self = false
+ to_port = 8080
+ },
+ {
+ cidr_blocks = [
+ "59.92.237.185/32",
+ ]
+ description = "SSH"
+ from_port = 22
+ ipv6_cidr_blocks = []
+ prefix_list_ids = []
+ protocol = "tcp"
+ security_groups = []
+ self = false
+ to_port = 22
+ },
+ ],
+ name = "jenkins-eks-demo-jenkins-sg"
+ name_prefix = (known after apply)
+ owner_id = (known after apply)
+ revoke_rules_on_delete = false
+ tags = {
+ "Name" = "jenkins-eks-demo-jenkins-sg"
+ }
+ tags_all = {
+ "Name" = "jenkins-eks-demo-jenkins-sg"
+ }
}

```

```

+ resource "aws_route_table" "public" {
+ arn = (known after apply)
+ id = (known after apply)
+ owner_id = (known after apply)
+ propagating_vgws = (known after apply)
+ route = [
+ {
+ cidr_block = "0.0.0.0/0"
+ gateway_id = (known after apply)
+ # (11 unchanged attributes hidden)
+ },
+ ],
+ tags = {
+ "Name" = "jenkins-eks-demo-public-rt"
+ }
+ tags_all = {
+ "Name" = "jenkins-eks-demo-public-rt"
+ }
+ vpc_id = (known after apply)
}

# aws_route_table_association.public will be created
+ resource "aws_route_table_association" "public" {
+ id = (known after apply)
+ route_table_id = (known after apply)
+ subnet_id = (known after apply)
}

# aws_security_group.jenkins will be created
+ resource "aws_security_group" "jenkins" {
+ arn = (known after apply)
+ description = "Allow SSH, Jenkins UI, and HTTP from my IP"
+ egress = [
+ {
+ cidr_blocks = [
+ "0.0.0.0/0",
+ ]
+ from_port = 0
+ ipv6_cidr_blocks = []
+ prefix_list_ids = []
+ protocol = "-1"
+ security_groups = []
+ self = false
+ to_port = 0
+ },
+ ],
+ }

```

```

+ ephemeral_block_device (known after apply)

+ instance_market_options (known after apply)

+ maintenance_options (known after apply)

+ metadata_options (known after apply)

+ network_interface (known after apply)

+ private_dns_name_options (known after apply)

+ root_block_device {
  + delete_on_termination = true
  + device_name           = (known after apply)
  + encrypted             = (known after apply)
  + iops                  = (known after apply)
  + kms_key_id            = (known after apply)
  + tags_all              = (known after apply)
  + throughput            = (known after apply)
  + volume_id             = (known after apply)
  + volume_size           = 20
  + volume_type           = "gp3"
}

# aws_internet_gateway.main will be created
+ resource "aws_internet_gateway" "main" {
  + arn          = (known after apply)
  + id           = (known after apply)
  + owner_id     = (known after apply)
  + tags        = {
    + "Name" = "jenkins-eks-demo-igw"
  }
  + tags_all    = {
    + "Name" = "jenkins-eks-demo-igw"
  }
  + vpc_id      = (known after apply)
}

# aws_route_table.public will be created
+ resource "aws_route_table" "public" {
  + arn          = (known after apply)

```

```

+ id                  = (known after apply)
+ instance_initiated_shutdown_behavior = (known after apply)
+ instance_lifecycle  = (known after apply)
+ instance_state      = (known after apply)
+ instance_type       = "t3.medium"
+ ipv6_address_count  = (known after apply)
+ ipv6_addresses      = (known after apply)
+ key_name            = "jenkins-1"
+ monitoring          = (known after apply)
+ outpost_arn         = (known after apply)
+ password_data       = (known after apply)
+ placement_group     = (known after apply)
+ placement_partition_number = (known after apply)
+ primary_network_interface_id = (known after apply)
+ private_dns         = (known after apply)
+ private_ip          = (known after apply)
+ public_dns          = (known after apply)
+ public_ip           = (known after apply)
+ secondary_private_ips = (known after apply)
+ security_groups     = (known after apply)
+ source_dest_check   = true
+ spot_instance_request_id = (known after apply)
+ subnet_id           = (known after apply)
+ tags               = {
  + "Name" = "jenkins-eks-demo-jenkins-server"
}
+ tags_all           = {
  + "Name" = "jenkins-eks-demo-jenkins-server"
}
+ tenancy            = (known after apply)
+ user_data           = "452e1b088b09d986984eaa1415cc61af16a54ed8"
+ user_data_base64   = (known after apply)
+ user_data_replace_on_change = false
+ vpc_security_group_ids = (known after apply)

+ capacity_reservation_specification (known after apply)

+ cpu_options (known after apply)

+ ebs_block_device (known after apply)

+ enclave_options (known after apply)

+ ephemeral_block_device (known after apply)

```

```
C:\Windows\System32\cmd.exe x + v
} + Version = "2012-10-17"
}
+ create_date = (known after apply)
+ force_detach_policies = false
+ id = (known after apply)
+ managed_policy_arns = (known after apply)
+ max_session_duration = 3600
+ name = "jenkins-eks-demo-jenkins-role"
+ name_prefix = (known after apply)
+ path = "/"
+ tags_all = (known after apply)
+ unique_id = (known after apply)
}
+ inline_policy (known after apply)
}

# aws_iam_role_policy_attachment.jenkins_admin will be created
+ resource "aws_iam_role_policy_attachment" "jenkins_admin" {
+ id = (known after apply)
+ policy_arn = "arn:aws:iam::aws:policy/AdministratorAccess"
+ role = "jenkins-eks-demo-jenkins-role"
}

# aws_instance.jenkins will be created
+ resource "aws_instance" "jenkins" {
+ ami = "ami-06033d1583f2e66ec"
+ arn = (known after apply)
+ associate_public_ip_address = (known after apply)
+ availability_zone = (known after apply)
+ cpu_core_count = (known after apply)
+ cpu_threads_per_core = (known after apply)
+ disable_api_stop = (known after apply)
+ disable_api_termination = (known after apply)
+ ebs_optimized = (known after apply)
+ enable_primary_ipv6 = (known after apply)
+ get_password_data = false
+ host_id = (known after apply)
+ host_resource_group_arn = (known after apply)
+ iam_instance_profile = "jenkins-eks-demo-jenkins-profile"
+ id = (known after apply)
+ instance_initiated_shutdown_behavior = (known after apply)
+ instance_lifecycle = (known after apply)
+ instance_state = (known after apply)
}
```

```
C:\Windows\System32\cmd.exe x + v

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.

C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\terraform>terraform apply
data.aws_availability_zones.available: Reading...
data.aws_ami.amazon_linux: Reading...
data.aws_availability_zones.available: Read complete after 0s [id=ap-south-1]
data.aws_ami.amazon_linux: Read complete after 0s [id=ami-06033d1583f2e66ec]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# aws_iam_instance_profile.jenkins_profile will be created
+ resource "aws_iam_instance_profile" "jenkins_profile" {
+ arn = (known after apply)
+ create_date = (known after apply)
+ id = (known after apply)
+ name = "jenkins-eks-demo-jenkins-profile"
+ name_prefix = (known after apply)
+ path = "/"
+ role = "jenkins-eks-demo-jenkins-role"
+ tags_all = (known after apply)
+ unique_id = (known after apply)
}

# aws_iam_role.jenkins_role will be created
+ resource "aws_iam_role" "jenkins_role" {
+ arn = (known after apply)
+ assume_role_policy = jsonencode(
+ {
+   Statement = [
+     {
+       Action = "sts:AssumeRole"
+       Effect = "Allow"
+       Principal = {
+         Service = "ec2.amazonaws.com"
+       }
+     }
+   ],
+   Version = "2012-10-17"
+ }
+ }
}
```

```
C:\Windows\System32\cmd.exe x + v
+ tags = {
+   + "Name" = "jenkins-eks-demo-public-subnet"
+ }
+ tags_all = {
+   + "Name" = "jenkins-eks-demo-public-subnet"
+ }
+ vpc_id = (known after apply)
}

# aws_vpc.main will be created
+ resource "aws_vpc" "main" {
+   arn = (known after apply)
+   cidr_block = "10.0.0.0/16"
+   default_network_acl_id = (known after apply)
+   default_route_table_id = (known after apply)
+   default_security_group_id = (known after apply)
+   dhcp_options_id = (known after apply)
+   enable_dns_hostnames = true
+   enable_dns_support = true
+   enable_network_address_usage_metrics = (known after apply)
+   id = (known after apply)
+   instance_tenancy = "default"
+   ipv6_association_id = (known after apply)
+   ipv6_cidr_block = (known after apply)
+   ipv6_cidr_block_network_border_group = (known after apply)
+   main_route_table_id = (known after apply)
+   owner_id = (known after apply)
+   tags = {
+     + "Name" = "jenkins-eks-demo-vpc"
+   }
+   tags_all = {
+     + "Name" = "jenkins-eks-demo-vpc"
+   }
+ }

Plan: 10 to add, 0 to change, 0 to destroy.

Changes to Outputs:
+ jenkins_public_ip = (known after apply)
+ jenkins_url = (known after apply)
+ vpc_id = (known after apply)
```

```
C:\Windows\System32\cmd.exe x + v
+ cidr_blocks = [
+   + "59.92.237.185/32",
+ ]
+ description = "SSH"
+ from_port = 22
+ ipv6_cidr_blocks = []
+ prefix_list_ids = []
+ protocol = "tcp"
+ security_groups = []
+ self = false
+ to_port = 22
+ },
+ name = "jenkins-eks-demo-jenkins-sg"
+ name_prefix = (known after apply)
+ owner_id = (known after apply)
+ revoke_rules_on_delete = false
+ tags = {
+   + "Name" = "jenkins-eks-demo-jenkins-sg"
+ }
+ tags_all = {
+   + "Name" = "jenkins-eks-demo-jenkins-sg"
+ }
+ vpc_id = (known after apply)
}

# aws_subnet.public will be created
+ resource "aws_subnet" "public" {
+   arn = (known after apply)
+   assign_ipv6_address_on_creation = false
+   availability_zone = "ap-south-1a"
+   availability_zone_id = (known after apply)
+   cidr_block = "10.0.1.0/24"
+   enable_dns64 = false
+   enable_resource_name_dns_a_record_on_launch = false
+   enable_resource_name_dns_aaaa_record_on_launch = false
+   id = (known after apply)
+   ipv6_cidr_block_association_id = (known after apply)
+   ipv6_native = false
+   map_public_ip_on_launch = true
+   owner_id = (known after apply)
+   private_dns_hostname_type_on_launch = (known after apply)
+   tags = {
+     + "Name" = "jenkins-eks-demo-public-subnet"
+   }
+ }
```

```
C:\Windows\System32\cmd.exe x + v
# aws_route_table_association.public will be created
+ resource "aws_route_table_association" "public" {
+   id = (known after apply)
+   route_table_id = (known after apply)
+   subnet_id = (known after apply)
+ }

# aws_security_group.jenkins will be created
+ resource "aws_security_group" "jenkins" {
+   arn = (known after apply)
+   description = "Allow SSH, Jenkins UI, and HTTP from my IP"
+   egress = [
+     {
+       cidr_blocks = [
+         "0.0.0.0/0",
+       ]
+       from_port = 0
+       ipv6_cidr_blocks = []
+       prefix_list_ids = []
+       protocol = "-1"
+       security_groups = []
+       self = false
+       to_port = 0
+     },
+     {
+       id = (known after apply)
+       ingress = [
+         {
+           cidr_blocks = [
+             "59.92.237.185/32",
+           ]
+           description = "Jenkins UI"
+           from_port = 8080
+           ipv6_cidr_blocks = []
+           prefix_list_ids = []
+           protocol = "tcp"
+           security_groups = []
+           self = false
+           to_port = 8080
+         },
+         {
+           cidr_blocks = [
+             "59.92.237.185/32",
+           ]
+           description = "Jenkins UI"
+           from_port = 8080
+           ipv6_cidr_blocks = []
+           prefix_list_ids = []
+           protocol = "tcp"
+           security_groups = []
+           self = false
+           to_port = 8080
+         }
+       ]
+     }
+   ]
+ }
```

```
C:\Windows\System32\cmd.exe x + v
+ volume_id = (known after apply)
+ volume_size = 20
+ volume_type = "gp3"
+ }
+ }

# aws_internet_gateway.main will be created
+ resource "aws_internet_gateway" "main" {
+   arn = (known after apply)
+   id = (known after apply)
+   owner_id = (known after apply)
+   tags = {
+     "Name" = "jenkins-eks-demo-igw"
+   }
+   tags_all = {
+     "Name" = "jenkins-eks-demo-igw"
+   }
+   vpc_id = (known after apply)
+ }

# aws_route_table.public will be created
+ resource "aws_route_table" "public" {
+   arn = (known after apply)
+   id = (known after apply)
+   owner_id = (known after apply)
+   propagating_vgws = (known after apply)
+   route = [
+     {
+       cidr_block = "0.0.0.0/0"
+       gateway_id = (known after apply)
+     },
+     {
+       id = (known after apply)
+       ingress = [
+         {
+           cidr_blocks = [
+             "59.92.237.185/32",
+           ]
+           description = "Jenkins UI"
+           from_port = 8080
+           ipv6_cidr_blocks = []
+           prefix_list_ids = []
+           protocol = "tcp"
+           security_groups = []
+           self = false
+           to_port = 8080
+         },
+         {
+           cidr_blocks = [
+             "59.92.237.185/32",
+           ]
+           description = "Jenkins UI"
+           from_port = 8080
+           ipv6_cidr_blocks = []
+           prefix_list_ids = []
+           protocol = "tcp"
+           security_groups = []
+           self = false
+           to_port = 8080
+         }
+       ]
+     }
+   ]
+   tags = {
+     "Name" = "jenkins-eks-demo-public-rt"
+   }
+   tags_all = {
+     "Name" = "jenkins-eks-demo-public-rt"
+   }
+   vpc_id = (known after apply)
+ }

# aws_route_table_association.public will be created
+ resource "aws_route_table_association" "public" {
+   id = (known after apply)
+   route_table_id = (known after apply)
+   subnet_id = (known after apply)
+ }
```

```
C:\Windows\System32\cmd.exe  +  v

+ spot_instance_request_id      = (known after apply)
+ subnet_id                    = (known after apply)
+ tags                         = {
+   + "Name" = "jenkins-eks-demo-jenkins-server"
+ }
+ tags_all                     = {
+   + "Name" = "jenkins-eks-demo-jenkins-server"
+ }
+ tenancy                      = (known after apply)
+ user_data                    = "452e1b08b09d986984eaa1415cc61af16a54ed8"
+ user_data_base64             = (known after apply)
+ user_data_replace_on_change  = false
+ vpc_security_group_ids       = (known after apply)

+ capacity_reservation_specification (known after apply)

+ cpu_options (known after apply)

+ ebs_block_device (known after apply)

+ enclave_options (known after apply)

+ ephemeral_block_device (known after apply)

+ instance_market_options (known after apply)

+ maintenance_options (known after apply)

+ metadata_options (known after apply)

+ network_interface (known after apply)

+ private_dns_name_options (known after apply)

+ root_block_device {
+   + delete_on_termination = true
+   + device_name           = (known after apply)
+   + encrypted             = (known after apply)
+   + iops                  = (known after apply)
+   + kms_key_id            = (known after apply)
+   + tags_all              = (known after apply)
+   + throughput            = (known after apply)
+   + volume_id             = (known after apply)
+   + volume_size           = 20
+ }
```

```
C:\Windows\System32\cmd.exe  +  v

+ id      = (known after apply)
+ policy_arn = "arn:aws:iam::aws:policy/AdministratorAccess"
+ role     = "jenkins-eks-demo-jenkins-role"
}

# aws_instance.jenkins will be created
+ resource "aws_instance" "jenkins" {
+   ami                      = "ami-06823d1583f2e66ec"
+   arn                     = (known after apply)
+   associate_public_ip_address = (known after apply)
+   availability_zone        = (known after apply)
+   cpu_core_count           = (known after apply)
+   cpu_threads_per_core     = (known after apply)
+   disable_api_stop         = (known after apply)
+   disable_api_termination  = (known after apply)
+   ebs_optimized            = (known after apply)
+   enable_primary_ipv6      = (known after apply)
+   get_password_data        = false
+   host_id                  = (known after apply)
+   host_resource_group_arn  = (known after apply)
+   iam_instance_profile     = "jenkins-eks-demo-jenkins-profile"
+   id                      = (known after apply)
+   instance_initiated_shutdown_behavior = (known after apply)
+   instance_lifecycle       = (known after apply)
+   instance_state           = (known after apply)
+   instance_type            = "t3.medium"
+   ipv6_address_count       = (known after apply)
+   ipv6_addresses           = (known after apply)
+   key_name                 = "jenkins-1"
+   monitoring               = (known after apply)
+   outpost_arn              = (known after apply)
+   password_data            = (known after apply)
+   placement_group          = (known after apply)
+   placement_partition_number = (known after apply)
+   primary_network_interface_id = (known after apply)
+   private_dns              = (known after apply)
+   private_ip               = (known after apply)
+   public_dns               = (known after apply)
+   public_ip                = (known after apply)
+   secondary_private_ips    = (known after apply)
+   security_groups          = (known after apply)
+   source_dest_check         = true
+   spot_instance_request_id = (known after apply)
+   subnet_id                = (known after apply)
+ }
```

```

+ id           = (known after apply)
+ name         = "jenkins-eks-demo-jenkins-profile"
+ name_prefix  = (known after apply)
+ path         = "/"
+ role         = "jenkins-eks-demo-jenkins-role"
+ tags_all     = (known after apply)
+ unique_id    = (known after apply)
}

# aws_iam_role.jenkins_role will be created
+ resource "aws_iam_role" "jenkins_role" {
+   arn           = (known after apply)
+   assume_role_policy = jsonencode(
+     { Statement = [
+       {
+         Action = "sts:AssumeRole"
+         Effect = "Allow"
+         Principal = {
+           Service = "ec2.amazonaws.com"
+         }
+       }
+     ],
+     Version = "2012-10-17"
+   }
+ )
+ create_date      = (known after apply)
+ force_detach_policies = false
+ id              = (known after apply)
+ managed_policy_arns = (known after apply)
+ max_session_duration = 3600
+ name            = "jenkins-eks-demo-jenkins-role"
+ name_prefix     = (known after apply)
+ path            = "/"
+ tags_all        = (known after apply)
+ unique_id       = (known after apply)
+ inline_policy (known after apply)
}

# aws_iam_role_policy_attachment.jenkins_admin will be created
+ resource "aws_iam_role_policy_attachment" "jenkins_admin" {
+   id           = (known after apply)
+   policy_arn   = "arn:aws:iam::aws:policy/AdministratorAccess"
}

```

4. Actual Terraform source files (vpc.tf, iam.tf, ec2.tf, variables.tf, providers.tf, outputs.tf, jenkins-userdata plus a Docker push

```

on windows_amd64

C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\terraform>terraform init
Initializing the backend...

Initializing provider plugins...
- Finding hashicorp/aws versions matching ">= 5.0"...
- Installing hashicorp/aws v5.100.0...
- Installed hashicorp/aws v5.100.0 (signed by HashiCorp)

Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.

C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\terraform>notepad terraform.tfvars
C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\terraform>notepad terraform.tfvars

C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\terraform>terraform plan
data.aws_ami.amazon_linux: Reading...
data.aws_availability_zones.available: Reading...
data.aws_availability_zones.available: Read complete after 0s [id=ap-south-1]
data.aws_ami.amazon_linux: Read complete after 1s [id=ami-06033d1583f2e66ec]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# aws_iam_instance_profile.jenkins_profile will be created
+ resource "aws_iam_instance_profile" "jenkins_profile" {
+   arn           = (known after apply)
+   create_date   = (known after apply)
}

```

```
C:\Windows\System32\cmd.exe
curl -LO "https://s3.us-west-2.amazonaws.com/amazon-eks/1.30.0/2024-05-12/bin/linux/amd64/kubectl"
chmod +x kubectl
mv kubectl /usr/local/bin/

# ---- Install eksctl ----
curl --silent --location "https://github.com/eksctl-io/eksctl/releases/latest/download/eksctl_Linux_amd64.tar.gz" | tar xz -C /tmp
mv /tmp/eksctl /usr/local/bin

# ---- Install Git ----
yum install -y git

# ---- Restart Jenkins after adding to docker group ----
systemctl restart jenkins
systemctl restart docker

echo "Jenkins, Docker, kubectl, eksctl installation complete." > /var/log/user-data-complete.log

C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\terraform>type outputs.tf
output "jenkins_public_ip" {
  description = "Public IP of the Jenkins EC2 instance"
  value       = aws_instance.jenkins.public_ip
}

output "jenkins_url" {
  description = "URL to access the Jenkins UI"
  value       = "http://${aws_instance.jenkins.public_ip}:8080"
}

output "vpc_id" {
  description = "ID of the created VPC"
  value       = aws_vpc.main.id
}

C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\terraform>terraform init
'terraform' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\terraform>where terraform
INFO: Could not find files for the given pattern(s).

C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\terraform>terraform -version
Terraform v1.16.1
on windows_amd64
```

```
C:\Windows\System32\cmd.exe

resource "aws_instance" "jenkins" {
  ami           = data.aws_ami.amazon_linux.id
  instance_type = var.instance_type
  subnet_id     = aws_subnet.public.id
  vpc_security_group_ids = [aws_security_group.jenkins.id]
  iam_instance_profile = aws_iam_instance_profile.jenkins_profile.name
  key_name       = var.key_name

  user_data = file("${path.module}/jenkins-userdata.sh")

  root_block_device {
    volume_size = 20
    volume_type = "gp3"
  }

  tags = {
    Name = "${var.project_name}-jenkins-server"
  }
}

C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\terraform>type jenkins-userdata.sh
#!/bin/bash
set -e

# ---- Update system ----
yum update -y

# ---- Install Java (required by Jenkins) ----
yum install -y java-17-amazon-corretto

# ---- Install Jenkins ----
wget -O /etc/yum.repos.d/jenkins.repo https://pkg.jenkins.io/redhat-stable/jenkins.repo
rpm --import https://pkg.jenkins.io/redhat-stable/jenkins.io-2023.key
yum install -y jenkins
systemctl enable jenkins
systemctl start jenkins

# ---- Install Docker ----
yum install -y docker
systemctl enable docker
systemctl start docker
usermod -aG docker jenkins
usermod -aG docker ec2-user
```

```
C:\Windows\System32\cmd.exe x + v

name = "${var.project_name}-jenkins-role"

assume_role_policy = jsonencode({
  Version = "2012-10-17"
  Statement = [
    {
      Action = "sts:AssumeRole"
      Effect = "Allow"
      Principal = {
        Service = "ec2.amazonaws.com"
      }
    }
  ]
})
}

# Broad access for a learning/demo environment.
# For production, replace with a tightly scoped custom policy.
resource "aws_iam_role_policy_attachment" "jenkins_admin" {
  role      = aws_iam_role.jenkins_role.name
  policy_arn = "arn:aws:iam::aws:policy/AdministratorAccess"
}

resource "aws_iam_instance_profile" "jenkins_profile" {
  name = "${var.project_name}-jenkins-profile"
  role = aws_iam_role.jenkins_role.name
}

C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\terraform>type ec2.tf
# ---- Latest Amazon Linux 2023 AMI ----
data "aws_ami" "amazon_linux" {
  most_recent = true
  owners      = ["amazon"]

  filter {
    name   = "name"
    values = ["al2023-ami-*-x86_64"]
  }

  filter {
    name   = "virtualization-type"
    values = ["hvm"]
  }
}

resource "aws_route_table_association" "public" {
  subnet_id      = aws_subnet.public.id
  route_table_id = aws_route_table.public.id
}

# ---- Security Group: Jenkins EC2 ----
resource "aws_security_group" "jenkins" {
  name        = "${var.project_name}-jenkins-sg"
  description = "Allow SSH, Jenkins UI, and HTTP from my IP"
  vpc_id      = aws_vpc.main.id

  ingress {
    description = "SSH"
    from_port   = 22
    to_port     = 22
    protocol    = "tcp"
    cidr_blocks = [var.my_ip_cidr]
  }

  ingress {
    description = "Jenkins UI"
    from_port   = 8080
    to_port     = 8080
    protocol    = "tcp"
    cidr_blocks = [var.my_ip_cidr]
  }

  egress {
    from_port   = 0
    to_port     = 0
    protocol    = "-1"
    cidr_blocks = ["0.0.0.0/0"]
  }

  tags = {
    Name = "${var.project_name}-jenkins-sg"
  }
}

C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\terraform>type iam.tf
# IAM role that the Jenkins EC2 instance assumes, so Jenkins can call
# ECR / EKS / EC2 APIs without static credentials on the box.
```

```
C:\Windows\System32\cmd.e  x  +  v

enable_dns_hostnames = true

tags = {
  Name = "${var.project_name}-vpc"
}
}

# ---- Internet Gateway ----
resource "aws_internet_gateway" "main" {
  vpc_id = aws_vpc.main.id

  tags = {
    Name = "${var.project_name}-igw"
  }
}

# ---- Public Subnet ----
resource "aws_subnet" "public" {
  vpc_id      = aws_vpc.main.id
  cidr_block  = var.public_subnet_cidr
  map_public_ip_on_launch = true
  availability_zone = data.aws_availability_zones.available.names[0]

  tags = {
    Name = "${var.project_name}-public-subnet"
  }
}

data "aws_availability_zones" "available" {
  state = "available"
}

# ---- Route Table ----
resource "aws_route_table" "public" {
  vpc_id = aws_vpc.main.id

  route {
    cidr_block = "0.0.0.0/0"
    gateway_id = aws_internet_gateway.main.id
  }

  tags = {
    Name = "${var.project_name}-public-rt"
  }
}

variable "aws_region" {
  description = "AWS region to deploy into"
  type        = string
  default     = "ap-south-1"
}

variable "project_name" {
  description = "Name prefix for resources"
  type        = string
  default     = "jenkins-eks-demo"
}

variable "vpc_cidr" {
  description = "CIDR block for the VPC"
  type        = string
  default     = "10.0.0.0/16"
}

variable "public_subnet_cidr" {
  description = "CIDR block for the public subnet"
  type        = string
  default     = "10.0.1.0/24"
}

variable "instance_type" {
  description = "EC2 instance type for the Jenkins server"
  type        = string
  default     = "t3.medium"
}

variable "key_name" {
  description = "Name of an existing EC2 key pair for SSH access"
  type        = string
}

variable "my_ip_cidr" {
  description = "Your IP address in CIDR form, for SSH/Jenkins UI access (e.g. 1.2.3.4/32)"
  type        = string
}

C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\terraform>type vpc.tf
# ---- VPC ----
resource "aws_vpc" "main" {
  cidr_block = var.vpc_cidr
```

```
C:\Windows\System32\cmd.exe
aee45f435e6c: Pushed
latest: digest: sha256:470591d33574cffe9dffa9721f3716d5f5a9802f4ff2ba2f71420a50b2b3ea8c size: 856

C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app>cd ..
C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app>mkdir terraform
A subdirectory or file terraform already exists.
C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app>cd terraform
C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\terraform>dir
Volume in drive C has no label.
Volume Serial Number is E43F-BECE

Directory of C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\terraform

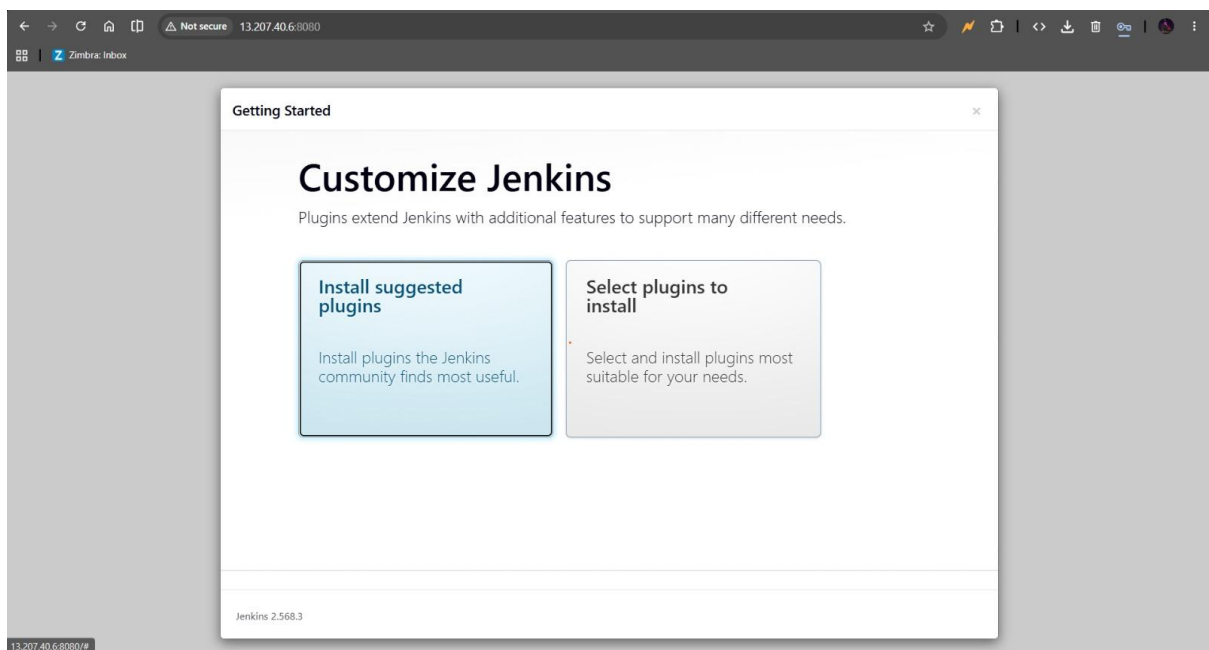
2026-09-06 18:22 <DIR>      .
2026-09-07 15:34 <DIR>      ..
2026-09-06 18:21           822 ec2.tf
2026-09-06 18:21           887 iam.tf
2026-09-06 18:21        1,170 jenkins-userdata.sh
2026-09-06 18:21           364 outputs.tf
2026-09-06 18:21        1,102 providers.tf
2026-09-06 18:21        1,819 vpc.tf
                6 File(s)          6,164 bytes
                2 Dir(s) 236,092,780,544 bytes free

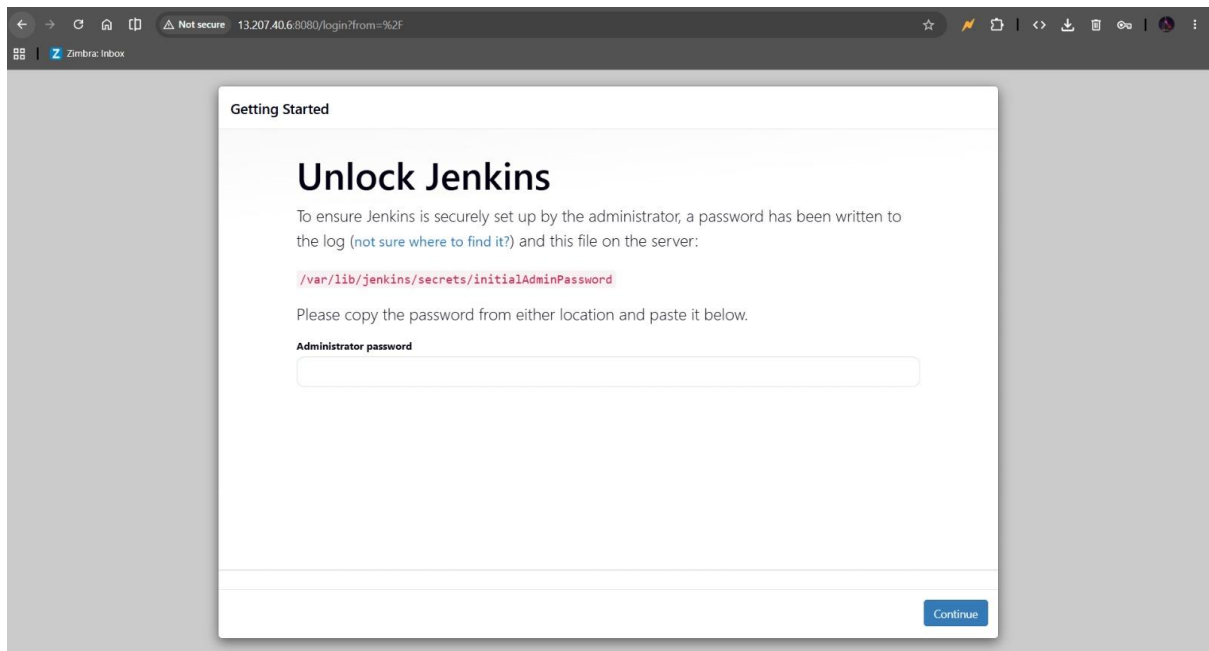
C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\terraform>type providers.tf
terraform {
  required_version = ">= 1.5.0"
  required_providers {
    aws = {
      source = "hashicorp/aws"
      version = "~> 5.0"
    }
  }
}

provider "aws" {
  region = var.aws_region
}

variable "aws_region" {
  description = "AWS region to deploy into"
```

5. Jenkins first-run setup screens (Un lock Jenkins, Customize Jenkins)





6. Docker login/ta g/push to Docker Hub, docker build, docker run, docker ps

```
C:\Windows\System32\cmd.e x + v

Password:
Login Succeeded

C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app>docker tag my-devops-app:latest thirulok2801/my-devops-app:latest

C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app>docker push thirulok2801/my-devops-app:latest
The push refers to repository [docker.io/thirulok2801/my-devops-app]
c74a82982b9d: Pushed
44136fa355b3: Mounted from docker/desktop-stats
dd71dde834b5: Pushed
f18232174bc9: Pushed
1e5a4c89cee5: Pushed
25ff2da83641: Pushed
4d171c336c5c: Pushed
8093f67ec345: Pushed
04e3230e2c6b: Pushed
aee45f435e6c: Pushed
latest: digest: sha256:470591d33574cffeefdf9721f3716d5f5a9802f4ff2ba2f71420a50b2b3ea8c size: 856

C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app>cd ..

C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app>mkdir terraform
A subdirectory or file terraform already exists.

C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app>
```

```
C:\Windows\System32\cmd.e x + v
C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app>docker stop test-run
test-run
C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app>docker rm test-run
test-run
C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app>docker login -u thirulok2001
Info -> A Personal Access Token (PAT) can be used instead.
To create a PAT, visit https://app.docker.com/settings
Password:
Error response from daemon: Get "https://registry-1.docker.io/v2/": unauthorized: incorrect username or password
C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app>docker login -u thirulok2001
Info -> A Personal Access Token (PAT) can be used instead.
To create a PAT, visit https://app.docker.com/settings
Password:
Error response from daemon: Get "https://registry-1.docker.io/v2/": unauthorized: incorrect username or password
C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app>docker login -u thirulok2001
Info -> A Personal Access Token (PAT) can be used instead.
To create a PAT, visit https://app.docker.com/settings
Password:
Login Succeeded
C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app>docker tag my-devops-app:latest thirulok2001/my-devops-app:latest
C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app>docker push thirulok2001/my-devops-app:latest
The push refers to repository [docker.io/thirulok2001/my-devops-app]
c74a02982b9d: Pushed
44136fa355b3: Mounted from docker/desktop-stats
dd71dde834b5: Pushed
```

```
C:\Windows\System32\cmd.e x + v
=> [internal] load build context 0.1s
=> transferring context: 889B 0.0s
=> [2/5] WORKDIR /app 2.5s
=> [3/5] COPY package*.json ./ 0.1s
=> [4/5] RUN npm install 4.8s
=> [5/5] COPY . . 0.1s
=> exporting to image 1.4s
=> exporting layers 0.6s
=> exporting manifest sha256:f84aa91e1beccfaa47862f30d0547a3031e9070c1744b39f9da3dd9b589f0e7 0.0s
=> exporting config sha256:98f5bb36bb8ab86d6e899ff45d3b01c87a25457b91faa7f594fa957ccl1f99e8 0.0s
=> exporting attestation manifest sha256:d1c56380b126138bd18523f42f16edc88e5463bc309b2a43aa8cb7eec93535e3 0.0s
=> exporting manifest list sha256:478591d33574cffe9dffa9721f3716d5f5a9802f4ff2ba2f71420a50b2b3ea8c 0.0s
=> naming to docker.io/library/my-devops-app:latest 0.0s
=> unpacking to docker.io/library/my-devops-app:latest 0.5s

C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app>docker images
IMAGE ID DISK USAGE CONTENT SIZE IN USE EXTRA
470024785907.dkr.ecr.ap-south-1.amazonaws.com/my-java-app:latest d37c780fd35e 299MB 88.5MB
docker/desktop-kubernetes:kubernetes-v1.36.1-cni-v1.9.1-critools-v1.35.0-cni-dockerd-v0.4.4-1-debian 65e5eddaacfc5 594MB 188MB
docker/desktop-stats:238597 dc25636ce73a 24.7MB 6.95MB U
docker/desktop-stats:238754 11882c43ecf4 24.7MB 6.95MB U
docker/desktop-storage-provisioner:v4.0 ca46f2a97ef4 82.1MB 24.7MB U
docker/desktop-vpnkit-controller:v4.0 8cb4e79ec634 52.6MB 11.9MB U
docker/docker-desktop-scout-desktop-module:latest c9bf22f5ee93 260MB 63.4MB
docker/welcome-to-docker:latest c4d56c24da4f 22.2MB 6.03MB U
my-devops-app:latest 470591d33574 197MB 47.5MB
my-java-app:latest d37c780fd35e 299MB 88.5MB
registry.k8s.io/coredns/coredns:v1.14.2 e7e6400cfd1e 182MB 22.5MB U
registry.k8s.io/etcd:3.6.8-0 39718941d1a 88.8MB 22.9MB U
registry.k8s.io/kube-apiserver:v1.36.1 2460df74307c 129MB 29.7MB U
registry.k8s.io/kube-controller-manager:v1.36.1 c8adb8daee5b 115MB 26.4MB U
registry.k8s.io/kube-proxy:v1.36.1 a96b6e12863e 111MB 28.9MB U
registry.k8s.io/kube-scheduler:v1.36.1 3a4333e8ba6e 83.4MB 20.4MB U
registry.k8s.io/pause:3.10.2 f548e0e8e3dc 1.06MB 318kB U

C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app>docker run -d -p 3000:3000 --name test-run my-devops-app:latest
e9e96c60ce096caf9d9dd47428aaf9806d2c3f6bc9598726d1d0af45a83a19bb

C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app>docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
e9e96c60ce00 my-devops-app:latest "docker-entrypoint.s..." 6 seconds ago Up 5 seconds 0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp test-run
```

7. Project init — git init, writing server.js/package.json/Dockerfile, git commit, git push to GitHub

```
C:\Windows\System32\cmd.exe x + v
2 Dir(s) 237,015,687,168 bytes free

C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app>ren Dockerfile.txt Dockerfile

C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app>dir
Volume in drive C has no label.
Volume Serial Number is E43F-BECE

Directory of C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app

2026-09-07 15:43 <DIR> .
2026-09-07 15:34 <DIR> ..
2026-09-07 15:42 45 .dockerignore
2026-09-07 15:37 26 .gitignore
2026-09-07 15:43 123 Dockerfile
2026-09-07 15:36 184 package.json
2026-09-07 15:35 375 server.js
5 File(s) 753 bytes
2 Dir(s) 237,015,252,992 bytes free

C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app>docker build -t my-devops-app:latest .
ERROR: failed to connect to the docker API at npipe:////./pipe/dockerDesktopLinuxEngine; check if the path is correct and if the daemon is running: open //.
/pipe/dockerDesktopLinuxEngine: The system cannot find the file specified.

C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app>docker build -t my-devops-app:latest .
[*] Building 21.4s (11/11) FINISHED
=> [internal] load build definition from Dockerfile
=> [internal] load metadata for docker.io/library/node:18-alpine 0.1s
=> transferring dockerfile: 1698B 0.0s
=> [internal] load metadata for docker.io/library/node:18-alpine 4.5s
=> [auth] library/node:pull token for registry-1.docker.io 0.0s
=> [internal] load .dockerignore 0.1s
=> transferring context: 858B 0.0s
=> [1/5] FROM docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9bc4b8c 7.3s
=> resolve docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9bc4b8c 0.1s
=> sha256:25ff2da83641908f65c3a74d80409d6b1b62ccfaab220b9ea70b80df5a2e0549 446B / 446B 0.4s
=> sha256:dd71dde834b5c203d162902e6b8994cb2309ae049a0eabc4feea161b2b5a3d0e 40.01MB / 40.01MB 3.8s
=> sha256:f18232174bc91741fdf3da96d85011092101a032a93a388b79e99e69c2d5c870 3.64MB / 3.64MB 5.6s
=> sha256:1e5a4c89cee5c0826c540ab06d4b6b491c96eda01837f430bd47f0d26702d6e3 1.26MB / 1.26MB 4.7s
=> extracting sha256:f18232174bc91741fdf3da96d85011092101a032a93a388b79e99e69c2d5c870 0.1s
=> extracting sha256:dd71dde834b5c203d162902e6b8994cb2309ae049a0eabc4feea161b2b5a3d0e 1.1s
=> extracting sha256:1e5a4c89cee5c0826c540ab06d4b6b491c96eda01837f430bd47f0d26702d6e3 0.1s
=> extracting sha256:25ff2da83641908f65c3a74d80409d6b1b62ccfaab220b9ea70b80df5a2e0549 0.0s
=> [internal] load build context 0.1s
=> transferring context: 889B 0.0s
```

```
C:\Windows\System32\cmd.exe x + v

C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app>git commit -m "Initial commit: sample node app"
[master (root-commit) 1c724cf] Initial commit: sample node app
3 files changed, 29 insertions(+)
create mode 100644 .gitignore
create mode 100644 package.json
create mode 100644 server.js

C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app>git remote add origin https://github.com/Thirulok007/my-devops-app.git

C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app>git branch -M main

C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app>git push -u origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 690 bytes | 345.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Thirulok007/my-devops-app.git
* [new branch] main -> main
branch 'main' set up to track 'origin/main'.

C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app>notepad .dockerignore

C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app>notepad Dockerfile

C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app>dir
Volume in drive C has no label.
Volume Serial Number is E43F-BECE

Directory of C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app

2026-09-07 15:43 <DIR> .
2026-09-07 15:34 <DIR> ..
2026-09-07 15:42 45 .dockerignore
2026-09-07 15:37 26 .gitignore
2026-09-07 15:43 123 Dockerfile.txt
2026-09-07 15:36 184 package.json
2026-09-07 15:35 375 server.js
5 File(s) 753 bytes
2 Dir(s) 237,015,687,168 bytes free
```

```
C:\Windows\System32\cmd.e  x  +  v
2026-09-07 12:11 <DIR> .
2026-09-07 12:02 <DIR> ..
                0 File(s)            0 bytes
                2 Dir(s)  237,254,148,096 bytes free

C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app>notepad server.js
C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app>
C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app>
C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app>notepad server.js
C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app>dir
Volume in drive C has no label.
Volume Serial Number is E43F-BECE

Directory of C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app

2026-09-07 15:34 <DIR> .
2026-09-07 15:34 <DIR> ..
2026-09-07 15:35                375 server.js
                1 File(s)            375 bytes
                2 Dir(s)  237,034,270,720 bytes free

C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app>notepad package.json
C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app>dir
Volume in drive C has no label.
Volume Serial Number is E43F-BECE

Directory of C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app

2026-09-07 15:36 <DIR> .
2026-09-07 15:34 <DIR> ..
2026-09-07 15:36                184 package.json
2026-09-07 15:35                375 server.js
                2 File(s)            559 bytes
                2 Dir(s)  237,033,103,360 bytes free

C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app>notepad .gitignore
C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app>git add .
C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app>git commit -m "Initial commit: sample node app"
```

```
C:\Windows\System32\cmd.e  x  +  v
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app>mkdir my-devops-app
A subdirectory or file my-devops-app already exists.

C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app>cd my-devops-app
C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app>git init
Reinitialized existing Git repository in C:/Users/DELL/OneDrive/Desktop/PROJECT 2/jenkins-terraform-eks-app/jenkins-terraform-eks-app/my-devops-app/.git/
C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app>git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    my-devops-app/

nothing added to commit but untracked files present (use "git add" to track)

C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app>dir
Volume in drive C has no label.
Volume Serial Number is E43F-BECE

Directory of C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app

2026-09-07 12:03 <DIR> .
2026-09-07 12:02 <DIR> ..
2026-09-07 12:03                my-devops-app
                0 File(s)            0 bytes
                3 Dir(s)  237,256,024,064 bytes free

C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app>rmdir my-devops-app
Access is denied.

C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app>rmdir /s /q my-devops-app
C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app>dir
Volume in drive C has no label.
Volume Serial Number is E43F-BECE

Directory of C:\Users\DELL\OneDrive\Desktop\PROJECT 2\jenkins-terraform-eks-app\jenkins-terraform-eks-app\my-devops-app
```